CSA4707-DEEP LEARNING

NAME:HARITHI S **REG NO:192421151**

Experiment 4: Entropy, Cross-Entropy and KL Divergence

Problem Statement: A weather-forecasting model outputs probabilities for Sunny, Cloudy and Rainy.

The team must measure how far its predictions are from what actually happened. Two competing forecast models are compared against 30 days of observed weather distributions for a city. The objective is to compute entropy, cross-entropy and KL divergence, and relate them to classification loss.

Use NumPy, SciPy, Matplotlib.

Tasks:

1. Compute the entropy of the observed weather distribution.
2. Compute cross-entropy and KL divergence of each model against the true distribution.
3. Verify that cross-entropy = entropy + KL divergence.
4. Show that KL divergence is asymmetric by computing it both ways.
5. Plot binary cross-entropy loss against predicted probability.

FORMULA:

1. Entropy: $H(P) = -\sum_i P_i \log_2 P_i$

2. Cross-entropy: $H(P, Q) = -\sum_i P_i \log_2 Q_i$

3. KL divergence: D_KL(P || Q) = Sum(P_i * log2(P_i / Q_i))

4. Verification: H(P, Q) = H(P) + D_KL(P || Q)

5. Binary cross-entropy: $L = -[y \ln(p) + (1 - y) \ln(1 - p)]$

PROGRAM CODE:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import entropy

P = np.array([0.5, 0.3, 0.2])
Q1 = np.array([0.6, 0.25, 0.15])
Q2 = np.array([0.4, 0.4, 0.2])

H = entropy(P, base=2)
CE1 = -np.sum(P * np.log2(Q1))
CE2 = -np.sum(P * np.log2(Q2))
KL1 = entropy(P, Q1, base=2)
KL2 = entropy(P, Q2, base=2)

print("Entropy:", round(H, 4))
print("Model 1 Cross-Entropy:", round(CE1, 4))
print("Model 1 KL Divergence:", round(KL1, 4))
print("Model 2 Cross-Entropy:", round(CE2, 4))
print("Model 2 KL Divergence:", round(KL2, 4))

print("Model 1 Verification:", round(H + KL1, 4))
print("Model 2 Verification:", round(H + KL2, 4))

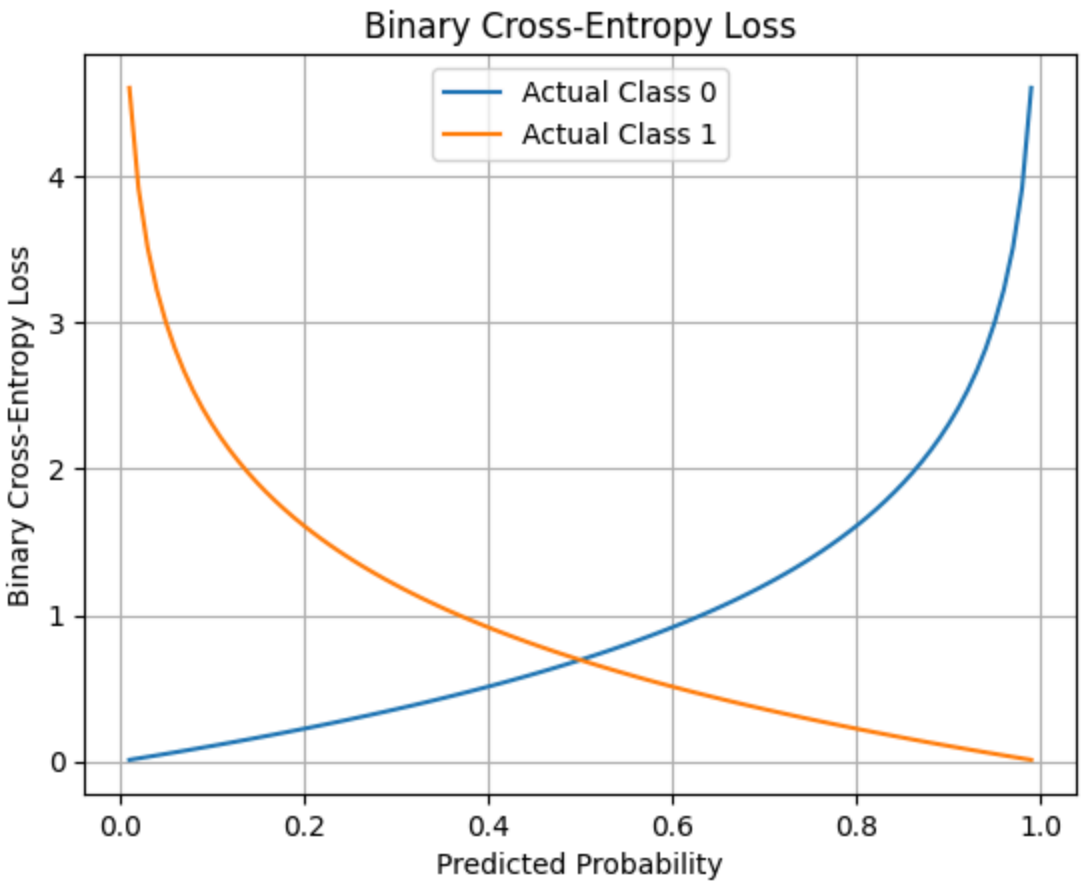
print("KL(P || Q1):", round(entropy(P, Q1, base=2), 4))
print("KL(Q1 || P):", round(entropy(Q1, P, base=2), 4))

p = np.linspace(0.01, 0.99, 100)
loss0 = -np.log(1 - p)
loss1 = -np.log(p)
```

```
plt.plot(p, loss0, label="Actual Class 0")
plt.plot(p, loss1, label="Actual Class 1")
plt.xlabel("Predicted Probability")
plt.ylabel("Binary Cross-Entropy Loss")
plt.title("Binary Cross-Entropy Loss")
plt.legend()
plt.grid()
plt.show()
```

OUTPUT:

```
Entropy: 1.4855
Model 1 Cross-Entropy: 1.5159
Model 1 KL Divergence: 0.0304
Model 2 Cross-Entropy: 1.5219
Model 2 KL Divergence: 0.0365
Model 1 Verification: 1.5159
Model 2 Verification: 1.5219
KL(P || Q1): 0.0304
KL(Q1 || P): 0.0298
```


Binary Cross-Entropy Loss
Actual Class 0
Actual Class 1
Binary Cross-Entropy Loss
0
1
2
3
4
0.0
0.2
0.4
0.6
0.8
1.0
Predicted Probability